Fraunhofer IAIS Research Paper Discussion

Scaling Synthetic Data for LLM Pre-Training:
Von Web-Filterung zu Instruction-Tuning

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Literaturverzeichnis & Zitationen

Paper 1: RefinedWeb

Penedo, G., Malartic, Q., Hesslow, D., Cojocaru, R., Cappelli, A., Alobeidli, H., Pannier, B., Almazrouei, E., and Launay, J. (2023). ‘The RefinedWeb Dataset for Falcon LLM: Outperforming Curated Corpora with Web Data, and Web Data Only’. arXiv:2306.01116. doi: 10.48550/arXiv.2306.01116

Paper 2: Nemotron-CC

Su, D., Kong, K., Lin, Y., Jennings, J., Norick, B., Kliegl, M., Patwary, M., Shoeybi, M., and Catanzaro, B. (2024). ‘Nemotron-CC: Transforming Common Crawl into a Refined Long-Horizon Pretraining Dataset’. arXiv:2412.02595. doi: 10.48550/arXiv.2412.02595

Paper 3: FineInstructions

Patel, A., Raffel, C., and Callison-Burch, C. (2025). ‘FineInstructions: Scaling Synthetic Instructions to Pre-Training Scale’. arXiv:2601.22146. doi: 10.48550/arXiv.2601.22146

Motivation: Das Data-Bottleneck Problem

LLM Training braucht Billionen von Tokens

Größere Datenmengen führen zu besserer Generalisierung, reduzieren Overfitting und ermöglichen es dem Modell, komplexere Muster zu lernen.

- **Llama 3.1** (8B–405B): 15T Tokens
- **Gemma 2** (27B): 13T Tokens

Problem: Aggressive Filterung entfernt Großteil der Daten

- FineWeb-Edu: 1.3T → **0.2T unique tokens** (80% Duplikate)
- DCLM: 3.8T → **1.0T unique tokens** (80% Duplikate)

Motivation: Das Data-Bottleneck Problem

Drei Paper, ein Ziel: Mehr nutzbare Daten

Pipeline-Evolution

Stage	Operation
CommonCrawl (Raw)	Input
RefinedWeb	Extract + Filter + Deduplicate
Nemotron-CC	+ Quality Buckets + Rephrasing
FineInstructions	+ Template Matching + Q&A Format

Research Goals: Zusammenhänge der drei Paper

Chronologische & konzeptionelle Progression

Paper	Jahr	Kernfrage	Ansatz
RefinedWeb	2023	Kann Web-Data allein mit Curated Data konkurrieren?	Aggressive Deduplizierung (Fuzzy + Exact)
Nemotron-CC	2025	Wie maximiert man Quality UND Quantity?	Classifier-Ensemble + Synthetic Rephrasing
FineInstructions	2026	Kann Pre-Training wie Instruction-Tuning aussehen?	Template-basierte Q&A-Generierung

Research Goals: Zusammenhänge der drei Paper

Gemeinsamer Nenner

- **CommonCrawl**
  - Wird von RefinedWeb und Nemotron explizit und FineInstruction implizit genutzt
- **Alle kämpfen gegen Duplikate**
  - Memorization vs. Generalization
- **Progression**: Minimal → Moderate → Maximal Transformation

Paper 1: RefinedWeb (Falcon LLM)

Penedo et al., 2023

Ausgangslage

Conventional Wisdom (vor 2023):

- Web-Daten < Curated Corpora (Books, Wikipedia, Papers)
- State-of-the-art: Mix aus Web (60%) + Curated (40%)
- Beispiel GPT-3: 300GT, davon 60% Web

Problem: Curation ist nicht skalierbar

- Llama 3.1 braucht 15T Tokens
- Größte öffentliche Datasets: ~1T Tokens (The Pile, C4, OSCAR)

Forschungsfrage: Kann **Web-only** mit Curated mithalten?

RefinedWeb: Methoden (MDR Pipeline)

MacroData Refinement: 3 Design-Prinzipien

1. **Scale First**: 3–6T Tokens Ziel (keine manuelle Curation)
2. **Strict Deduplication**: Fuzzy (MinHash) + Exact (Suffix Array)
3. **Neutral Filtering**: Nur Heuristiken, kein ML-Filter (Bias-Vermeidung)

Pipeline-Stufen

1. **Document Prep**: Trafilatura → FastText LangID (threshold 0.65)
2. **Filtering**: Repetition removal + Document/Line-wise heuristics
3. **Deduplication**: MinHash (9K hashes, 20 buckets) + Exact (≥50 tokens)

RefinedWeb: Deduplication Details

Warum so aggressiv?

- **Memorization-Problem**: Duplikate bei 175B Modellen schädlich (Hernandez et al., 2022)
- **MinHash-Konfiguration**: 9K hashes (vs. The Pile: 10 hashes)
  - 5-grams, 20 buckets à 450 hashes
  - Removal Rate: ~50% der Daten

Exact Substring Deduplication

Methode: Suffix Array → ≥50 Token Matches → Remove/Mask/Drop

- Suffix Arrays über konkatenierte Dokumente:
  - Jede Textposition wird zum Suffix-Startpunkt

RefinedWeb: Experimente & Results

Models trained on RefinedWeb EB alone outperform models trained on curated corpora

RefinedWeb: Experimente & Results

Key Findings

Small-Scale (1B @ 27GT, 3B @ 60GT):

- RefinedWeb > C4 > The Pile auf MMLU
- C4 stark, aber RefinedWeb +2–3% besser

Full-Scale (7B @ 350GT):

- **RefinedWeb matches GPT-3** (same eval setup)
- Open models (OPT, Pythia) underperform
- **Web-only funktioniert!**

RefinedWeb: Ablations & Takeaways

Ablation: Do Other Corpora Benefit from MDR?

Although improvements from filtering are not systematic across datasets, deduplication brings a steady performance boost across the board

Dataset	Base MMLU	+Filtered	+Dedup	+Both
OSCAR-21.09	55.0%	+0.4	+0.6	+0.5
C4	55.7%	+0.5	+0.2	+0.7
The Pile	**53.4%**	**+0.8**	**+1.1**	**+1.8**
RefinedWeb	52.7%	+1.6	-	+3.5

Erkenntnis: Deduplication hilft **konsistent**, Filtering ist dataset-spezifisch

RefinedWeb: Ablations & Takeaways

Limitations

- Toxicity ≈ The Pile (Perspective API)
- Multiple Epochs: Unklar ob deduplizierte Daten mehr Epochs erlauben
- Pythia-Studie: Dedup auf The Pile hatte **wenig Impact** (Widerspruch?)

Paper 2: Nemotron-CC

Su et al., 2024

Ausgangslage

RefinedWeb-Problem: ~90% Gesamtreduktion (alle Pipeline-Stufen kombiniert) → Bottleneck für 15T Training

- DCLM/FineWeb-Edu: ~90% Removal, nur 1T unique tokens
- **Trade-off**: Quality vs. Quantity

Forschungsfrage: Besserer Trade-off durch:

1. Classifier Ensembling (mehr Recall)
2. Synthetic Rephrasing (mehr Unique Tokens)
3. Weniger Heuristic Filters (mehr Yield)

Paper 2: Nemotron-CC

MMLU scores for 8B parameter models trained for 1T tokens.

Nemotron-CC: Methoden (3-Komponenten-Ansatz)

1. HTML-to-Text Extractor & Filter

Extractor-Wahl: Justext > Trafilatura (+28.6% HQ tokens)

Filter-Strategie: - **Heuristic Filters NUR auf Low-Quality Data** - High-Quality Data: Unfiltered (vermeidet 18.1% Token-Verlust)

Ablation studies on extractor and filter

Experiment	MMLU	Avg (9 tasks)
Trafilatura filtered	55.4	60.6
Justext filtered	54.1	60.9
Justext unfiltered	55.5	60.3
Justext HQ-unfiltered	**57.5**	**60.6**

Nemotron-CC: Model-Based Quality Labeling

Classifier-Ensemble (3 Modelle)

Klassifikatoren:

1. Mistral 8x22B-annotiert (Educational Value 0–5)
2. Nemotron-340B-annotiert (Educational Value 0–5)
3. DCLM FastText (Instruction + ELI5 Reddit)

Ensemble-Strategie: Maximum-Operation → 20 Buckets → 5 Quality Labels

Nemotron-CC: Model-Based Quality Labeling

High-Quality Documents Overlap Analysis

Kategorie	Dokumente	Total unique %	Erklärung
Intersection (alle 3 Classifier)	1.15M	10.1%	Dokumente, die von ALLEN 3 Klassifikatoren als High-Quality bewertet wurden
FineWeb-Edu only	4.02M	35.4%	Nur Mistral 8x22B + Nemotron-340B stimmen überein (DCLM weicht ab)
DCLM only	6.18M	54.4%	Nur DCLM FastText bewertet als High-Quality (andere Klassifikatoren nicht)

Common Crawl quality labels statistics

Label	Buckets	Tokens (B)	%
High	19	553	12.6%
Medium-High	18	504	11.5%
Medium	12–17	2,023	46.2%

Nemotron-CC: Synthetic Data Generation

Strategie: Low-Quality vs. High-Quality

Low-Quality (403B → 336B):

- **Ziel**: Noise/Error Reduction
- **Methode**: Wikipedia-Style Rephrasing
- **Resultat**: -16.6% (Qualitätskontrolle)

High-Quality (451B → 1.5T):

- **Ziel**: Token Diversity + Knowledge Condensation
- **Methode**: 5 Prompts (QA, Distill, Extract, List, Wikipedia)
- **Resultat**: +232% (Unique Token Variants)

→ Verhindert Overfitting durch Repetition (Muennighoff et al.)

Nemotron-CC: Synthetic Data Generation (2)

Synthetic data token count statistics

Prompt	Tokens	Beschreibung
Diverse QA Pairs	499.5B	Yes/No, Open-ended, Multi-choice
Distill	157.6B	Concise passage
Extract Knowledge	303.6B	Facts only, discard filler
Knowledge List	203.2B	Organized bullet points

Post-Processing: Remove incomplete, strip Markdown, filter <50 tokens

Nemotron-CC: Results

Short Horizon (1T tokens, 8B model)

Results for 8B parameter models trained on 1T tokens

Dataset	MMLU	ARC-C	Avg (10 tasks)
Nemotron-CC-HQ	**59.0**	**52.9**	**60.1**
DCLM	53.4	47.0	57.0
Nemotron-CC	53.0	50.7	57.8
FineWeb-Edu	42.9	48.0	53.2

Erkenntnis: HQ-Subset +5.6 MMLU, Full dataset = DCLM aber 4× mehr Daten

Nemotron-CC: Results

Long Horizon (15T tokens, 8B model)

Comparison of our 8B parameter model vs Llama 3.1 8B

Model	MMLU	ARC-C	Avg (10 tasks)
Ours (7.2T from Nemotron-CC)	**70.3**	**58.1**	**64.7**
Llama 3.1 8B	65.3	55.0	64.2

Erkenntnis: State-of-the-art über langen Horizont

Nemotron-CC: Ablations

Classifier Comparison

Different classifiers comparison

Classifier	HQ %	MMLU	Avg (9 tasks)
FineWeb-Edu	8%	55.4	59.0
DCLM	11%	56.0	58.4
Ours-Ensembled	**25%**	**56.4**	**59.4**

Erkenntnis: Ensemble verdoppelt HQ-Recall, hält Performance

Nemotron-CC: Ablations

Synthetic Data Impact

Impact of incorporating synthetic data

Blend	MMLU	ARC-E	Avg
LQ-Base	48.2	67.7	52.5
LQ-Synthetic	**47.1**	**71.3**	**54.0** (+1.5)
HQ-Base (8 epochs)	53.4	74.2	55.8
HQ-Synthetic (4 epochs real + 4 synthetic)	**53.6**	**76.7**	**56.7** (+0.9)

Erkenntnis: Rephrasing hilft, Fresh Tokens > Repeated Tokens

Paper 3: FineInstructions

Patel, Raffel, Callison-Burch, 2026

Ausgangslage

Pre-Training Misalignment:

- Standard: Self-supervised “predict next word” auf unstrukturiertem Text
- Instruction-Tuning: Supervised auf wenigen 1000–100K Beispielen
- **Problem**: Pre-Training-Objective ≠ Downstream Usage (User prompts)

Existing Instruction Datasets:

- Klein (few thousand) oder schmal (academic NLP tasks)
- LLM-generiert → Distillation, kein echtes Wissen

Paper 3: FineInstructions (2)

Forschungsfrage: Kann Pre-Training **direkt** als Instruction-Tuning erfolgen?

- **Idee**: Pre-Training-Dokumente → Instruction-Answer-Paare (1B+ Paare)

FineInstructions: Methoden (Pipeline)

4-Stufen-Prozess

1. Instruction Template Generation (~18M Templates)

Input: ~18M User-Queries (WildChat, LMSys, Reddit QA, GooAQ, etc.)

Prozess:

Schritt	Beschreibung
LLM	Llama-3.3 70B generiert 50K “Silver-Standard” Templates
Format	`"Between <fi>Entity A</fi> and <fi>Entity B</fi> which is more <fi>characteristic</fi>?"`
Distillation	Llama-3.2 1B trainiert auf 50K → generalisiert auf 18M

Output: 18M Templates + “Compatible Document Descriptions”

FineInstructions: Methoden (Pipeline)

2. Template ⇔ Document Matching

Embedding Model: BGE-M3 (fine-tuned 2×)

- **Round 1**: Hard positives/negatives (LLM-judged compatibility)
- **Round 2**: Gaussian Pooling Layer (K=5 chunks per document)

FineInstructions: Gaussian Pooling (Technical Detail)

Problem: Long Documents

- Global mean pooling → verliert lokale Information
- Templates relevant für **spezifische Abschnitte**, nicht ganzes Dokument

Lösung: K=5 Local Embeddings + 1 Global

Parameter: K=5, α=1.0 (pure local), σ=0.05

Validation: Pearson correlation 0.99 zwischen Chunk-Index und Answer-Location

Retrieval: Cosine Similarity > 0.865 → Match

FineInstructions: Instantiation & Judging

3. Instruction & Answer Generation

Instantiator Model: Llama-3.2 3B (distilled from 70B)

Groundedness: ≥80% der Antwort = Direct Excerpt aus Dokument

Compute Optimization: Excerpt Tags

- Original: “It is known that no preferred inertial frame exists according to the principle of relativity”
- Generated: “It is known that <…> the principle of relativity.”

Training: 2-Round Distillation

- Round 1: 100K Silver-Standard (Llama-3.3 70B)
- Round 2: 100K filtered by LLM-as-Judge (stratified: length, complexity, topics)

FineInstructions: Instantiation & Judging

4. Quality Filtering

Judge: Flow Judge (3.8B, 5-point Likert: 1=irrelevant, 5=perfect)

- Threshold: ≥4
- **Final Output**: 1B+ Instruction-Answer Pairs

FineInstructions: Methoden (Pipeline)

The FineInstructions pipeline for efficiently generating diverse, pre-training scale, synthetic instruction-answer pairs.

FineInstructions: Experiments

Setup

Baselines:

- **IPT** (Instruction Pre-Training): 23B tokens, Q&A from academic NLP
- **Nemotron-CC**: 300B tokens (Q&A, WRAP, Full Mix)

Models: 1.8B parameters (Llama-3 tokenizer, Lingua framework)

Data Blend: 73% CC (varied) + 27% fixed (code, papers, books)

Evaluation: 3 Benchmarks

FineInstructions: Experiments (2)

Benchmarks

- **MixEval**: Academic tasks (MMLU, TriviaQA, etc.) + LLM-as-Judge (GPT-5 mini)
- **MT-Bench-101**: Realistic user queries, 10-point Likert
- **AlpacaEval**: Head-to-head, length-bias corrected (GPT-4-Turbo)

FineInstructions: Results

1.8B parameter models across two pre-training corpora

Corpus	Method	MixEval Std	MT-Bench	AlpacaEval
IPT 23B	Standard Pre-Training	17.8	1.9	+47.2
IPT 23B	IPT	19.8	2.4	+36.4
IPT 23B	FineInstructions	31.7	2.8	Ref.
Nemotron 300B	Standard Pre-Training	24.0	3.5	+27.2
Nemotron 300B	WRAP	22.8	3.6	+30.2
Nemotron 300B	Nemotron Q&A	27.1	3.4	+52.2
Nemotron 300B	Nemotron-CC Full	24.5	3.6	+31.8
Nemotron 300B	FineInstructions	33.0	3.9	Ref.

AlpacaEval: Win margin for FineInstructions (higher = better)

FineInstructions: Diversity Analysis

Template Diversity

- **4.3M unique templates** verwendet (aus 18M Pool)
- Kein Template >0.09% der Daten

Quellen-Verteilung:

- GooAQ: ~50%
- Reddit QA: ~27%
- LMSys Chat: ~9%
- WildChat: ~6%

FineInstructions: Diversity Analysis

A visualization of the task diversity in FineInstructions

FineInstructions: Ablation (Judging Impact)

Incorporating a judging and filtering stage on top of FineInstructions leads to further improvements in performance

IPT Corpus (23B)

Method	MixEval Std	AlpacaEval Win % (vs. Judged FI)
FI (No Judging)	30.1	46.5% (Δ -7%)
FI (Judged)	**31.7**	—

Nemotron-CC Corpus (300B)

Method	MixEval Std	AlpacaEval Win % (vs. Judged FI)
FI (No Judging)	33.3	61.4% (Δ +22.8%)
FI (Judged)	**33.0**	—

Discussion: Methodische Schwachstellen

Kritikpunkt 1: Entity Replacement (Reproduzierbarkeit)

Problem:

- Paper: “Queries are genericized by replacing spans with `<fi></fi>` tags”
- **Keine Erklärung**: Wie werden Entity-Boundaries erkannt?
- **Keine Heuristiken**: Token-Classification? NER? Regelbasiert?

Konsequenz:

- Template-Generierung = **Black Box** (LLM-Prompts)
- Nur 1 Beispiel-Prompt in Appendix C
- **Nicht reproduzierbar** ohne Llama-3.3 70B + exakte Prompts

Discussion: Methodische Schwachstellen (3)

Kritikpunkt 2: Vague Prompting Methodology

Problem:

- Erstellung des silver standard
- “Prompting an LLM with a series of prompts” (Section 3.1)
- **Keine Details** zu:
  - Few-Shot Examples? (Anzahl, Auswahl)
  - Chain-of-Thought? (Reasoning Steps)
  - Multi-Step Verification? (Self-Consistency)
  - Temperature/Top-p Settings?

Discussion: Methodische Schwachstellen (4)

Konsequenz:

- **50K Silver-Standard Templates** = Foundation für gesamte Pipeline
- Query Genericizer (Llama-3.2 1B) hängt davon ab
- **Nicht reproduzierbar** ohne Prompt-Engineering-Details

Discussion: Statistische Validität

Kritikpunkt 3: Pearson Correlation (Gaussian Pooling)

Claim (Section 3.2):

“Pearson correlation of 0.99 between chunk index and answer location”

Problem:

- **Chunk Index** = diskrete Ordinalvariable (1, 2, 3, 4, 5)
- **Answer Location** = kontinuierlich (0–100% im Dokument)
- **Pearson-Annahme**: Normalverteilung, lineare Beziehung

Frage:

- Ist **Spearman Rank Correlation** nicht robuster?
- Non-parametrisch, keine Normalverteilungs-Annahme

Eigene Forschung: Parallelen zu FineInstructions

Masterarbeit (2017): Maschinelle Analyse sprachlicher Muster in Risikoberichten - Konzeptionierung und Evaluation inguistischer Modelle zur Beschreibung wirtschaftlicher Entwicklungen

Konzeptionelle Übereinstimmung

Konzept	Masterarbeit (2017)	FineInstructions (2025)
Template-Abstraktion	Phraseframes mit Slots (XXX)	Instruction Templates (`<fi></fi>`)
Slot-Filling	Automatisch (Korpus-basiert)	Automatisch (Dokument-Matching)
Matching	Signifikanz (Log-Likelihood)	Embeddings (BGE-M3, Cosine >0.865)
Clustering	Levenshtein-Distanz (≤2)	Semantic Similarity
Weak Supervision	Implizit (statistisch)	Explizit (Silver-Standard via LLM)

Beispiel Phraseframe: XXX consider XXX following risk factors

Quantitativer Vergleich

Metrik	Masterarbeit (2017)	FineInstructions (2025)
Templates/Frames	152K signifikante Frames → 24.654 Cluster	18M Templates → 1B+ Instruction-Paare
Dokumente	4.456 Risikoberichte (478 Unternehmen × 10 Jahre)	~100K Pre-Training-Dokumente
Erfolgsrate	81/478 Unternehmen (16.9%) mit signifikanter Korrelation (Return on no_doc/assetsts)	MMLU +5.6 vs. DCLM (1T), +5.0 vs. Llama 3.1 (15T)
Reproduzierbarkeit	Vollständige Perl-Skripte, dokumentierte Algorithmen	LLM-Prompts unvollständig, Entity-Replacement undokumentiert
Skalierungs-Strategie	Levenshtein-Clustering (Edit-Distanz ≤2)	Distillation (70B → 1B/3B) + FAISS-Index

	Masterarbeit (2017)	FineInstructions (2025)
Ansatz	Regelbasiert, transparent	LLM-basiert, Black Box
Performance	Rechenintensiv	Skalierbar

Lessons Learned: Die drei Papers im Vergleich

Dimension	RefinedWeb	Nemotron-CC	FineInstructions
Kernbeitrag	Web-only = Curated	Quality + Quantity	Pre-Training = Instruction-Tuning
Daten-Transformation	Minimal (Extract + Dedup)	Moderat (Rephrasing + Buckets)	Maximal (Q&A Format)
Skalierung	5T tokens (600B public)	6.3T tokens (4.4T real + 1.9T synthetic)	1B+ Instruction-Paare
Reproduzierbarkeit	Algorithmen dokumentiert	Heuristiken + LLM-Prompts	LLM Black Box
Stärke	Aggressive Deduplication	Classifier-Ensemble	Template Diversity
Schwäche	Hohe Removal Rate (77%)	Synthetic Data Hallucinations?	Undokumentierte Prompts

Takeaways für die Praxis

1. Data Engineering > Model Architecture

- **RefinedWeb**: Web-only schlägt Curated (bei richtiger Verarbeitung)
- **Nemotron-CC**: Ensemble-Filtering verdoppelt HQ-Recall
- **FineInstructions**: Format-Alignment (Q&A) verbessert Downstream-Performance

2. Skalierung

- **Qualität**: Aggressive Filterung (FineWeb-Edu: 90% Removal)
- **Quantität**: Weniger Filter + Synthetic Augmentation (Nemotron-CC: 4× mehr Daten)
- **Alignment**: Template-Matching (FineInstructions: Pre-Training = Instruction-Tuning)

Danke für die Aufmerksamkeit

Details und Code zur Präsentation finden Sie unter:

github.com/Scrottz/fraunhofer_research_paper_analysis

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